

Ömrüm Ceren GÜLER

Software Engineering Graduate

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Motivated to learn how to write error-free code for a large company, passionate about advancing in the software development field.

Education

- 1) **Eastern Mediterranean University**, Famagusta/Cyprus - Bachelor's Degree in Software Engineering
 - Graduate date: 07/2026
 - ABET accredited university and software engineering program
 - 100% English medium of instruction
- 2) Private Adana Ceyhan Final Anatolian High School, class of 2022 (GPA: 98/100)

Skills

- Python Programming Language
- C Programming Language
- Java
- Object Oriented Programming
- Ongoing Coursework: Client-Server Programming (HTML/CSS/Web Systems)
- SQL
- Database Management Systems
- Emerging Technologies
- Tinkercad
- MatLab
- Numerical Analysis
- Linear Algebra
- Data Structures
- Operating Systems
- Digital Logic Design
- Algorithm Analysis
- Discrete Mathematics
- Ordinary Differential Equations
- Probability and Statistical Methods
- Principles of Programming Languages
- Computer Networks

- Ethics in Engineering
- Software Requirements Analysis and Specification
- Automata Theory
- Software Process and Management
- Embedded System Design
- Data Science
- Artificial Intelligence
- English (Professional Proficiency)
- Turkish (Native)
- German (Beginner Level)

Internship Projects

Çukurova University, IT Department - Software Engineering Summer Intern
Adana/Turkey (July 2025 - September 2025)

1) **FastAPI Data Analytics & Prediction Service** | Backend Developer

Internship Project – Çukurova University

- **Project Overview:** Developed a FastAPI-based service application for Çukurova University to provide data analysis, predictive modeling, and various API endpoints.
- **Technical Implementation:** Established secure connections to SQL Server databases using SQLAlchemy and pyodbc for robust data retrieval.
- **Analytics & Forecasting:** Integrated Prophet and Linear Regression models to perform advanced data analysis and forecasting tasks.
- **System Management:** Implemented Swagger (OpenAPI) for interactive API testing and utilized .env files for secure management of sensitive database credentials.
- **Tools Used:** FastAPI, SQL Server, SQLAlchemy, Prophet, Linear Regression, Swagger UI.

2) **Automated Passport Photo Compliance System** | Computer Vision Developer

Internship Project – Çukurova University

- **Project Overview:** Built an automated API and bulk control tool to evaluate the compliance of ID and passport photos with official government standards.
- **Computer Vision Logic:** Leveraged MediaPipe and OpenCV for face detection, background analysis, and meticulous checks for contrast, color, and lighting.
- **Validation Features:** Engineered modules to verify image resolution, aspect ratios, EXIF data, and rotation, including optional accessory detection (glasses, hats, etc.).
- **Scalability & Security:** Developed a dedicated CLI client for bulk testing and implemented rate limiting to ensure API security and stability.
- **Tools Used:** FastAPI, MediaPipe, OpenCV, Python (CLI Client), Image Processing Techniques.

3) **Reservation Segmentation & Analytics System** | Data Analyst & Developer

Internship Project – Çukurova University

- **Project Overview:** Developed a Python-based analytical system to process corporate reservation data stored in SQL Server, dividing them into meaningful segments for business insights.

- Machine Learning Implementation: Applied the K-Means clustering algorithm to automatically group reservations into short-term and long-term categories based on calculated durations.
- Predictive Classification: Implemented a Random Forest classification model to predict the segment of new reservation entries based on time and location data.
- Data Engineering & Analysis: Performed time-based density analysis to identify peak hourly and daily reservation patterns and utilized Pandas for complex duration calculations.
- Visualization & Security: Generated statistical distribution charts and visual summaries using Matplotlib and secured database access credentials using .env files.
- Tools Used: Python, SQL Server, Scikit-Learn (K-Means, Random Forest), Pandas, PyODBC, Matplotlib, python-dotenv.

4) **Academic Publication Analysis & Forecasting System** | Data Analyst & Developer

Internship Project – Çukurova University

- Project Overview: Developed an analytical system for Çukurova University to evaluate academic publication trends and generate data-driven forecasts for the 2025 academic year.
- Statistical Modeling: Conducted rigorous model comparison studies between Linear Regression and Simple Average methods to determine optimal forecasting accuracy for unit-based and author-based metrics.
- Data Engineering: Executed complex data extraction and processing from SQL Server relational databases, managing multi-table schemas including author records and research area attributes.
- Analytical Insights: Identified top-performing academic units and the fastest-growing research domains by analyzing past publication trends and unit-specific popularity.
- Visualization & Security: Engineered automated reporting modules using ggplot2 and showtext for high-quality visual summaries, while securing database credentials through dotenv.
- Tools Used: R (dplyr, ggplot2, DBI, odbc, dotenv, showtext), SQL Server.

5) **Academic Publication Prediction & Analysis System** | Data Scientist & Developer

Internship Project – Çukurova University

- Project Overview: Developed a specialized analytical system to forecast future academic publication counts by research area, leveraging historical multi-year datasets.
- Forecasting Strategies: Implemented dual-modeling approaches using Facebook Prophet for advanced time-series forecasting and Linear Regression (utilizing numpy.polyfit) for comparative trend analysis.
- Data Engineering: Engineered direct data extraction pipelines from SQL Server utilizing SQLAlchemy for efficient retrieval and Pandas for complex data processing.
- Interactive Interface: Designed an interactive terminal menu allowing users to filter data by academic field, select specific prediction years, and switch between forecasting models dynamically.
- Visual Analytics: Automated the generation of graphical outputs using Matplotlib and Seaborn to visualize yearly publication trends and predictive comparisons.
- Security & Environment: Utilized python-dotenv to securely manage environment variables and sensitive database connection credentials.
- Tools Used: Python (3.8+), Facebook Prophet, Numpy, Pandas, SQLAlchemy, Matplotlib, Seaborn, SQL Server.

6) **Academic Success Classification & Performance Analysis System** | Data Scientist & Developer

Internship Project – Çukurova University

- **Project Overview:** Developed a comprehensive performance analysis system for Çukurova University to evaluate and classify the achievement levels of academic units using the STRATEJİK_PLAN database.
- **Data Science Workflow:** Executed a full-cycle data pipeline including direct data extraction from SQL databases, rigorous preprocessing, and classification logic based on targeted versus actual performance ratios.
- **Security & Privacy:** Implemented industry-standard security practices by managing database credentials through .env files and ensuring data privacy by utilizing anonymized datasets for testing.
- **System Architecture:** Designed a clean and maintainable project structure utilizing .gitignore for credential protection and requirements.txt for robust dependency management.
- **Automated Testing:** Developed specialized Python scripts for generating synthetic data samples to validate classification logic and system stability without compromising sensitive information.
- **Tools Used:** Python, SQL, Pandas, NumPy, python-dotenv, SQL Server.

7) **YÖKSİS Unit Hierarchy Visualization** | Data Visualization Developer

Internship Project – Çukurova University

- **Project Overview:** Developed an interactive treemap visualization tool to map the organizational hierarchy (faculties, institutes, departments) of Çukurova University using official YÖKSİS datasets.
- **Interactive Interface:** Engineered a dynamic dashboard using Streamlit and Plotly Express, enabling users to navigate seamlessly through multi-level hierarchies (Faculty Department Sub-unit).
- **Data Exploration Features:** Implemented advanced filtering logic including real-time search bars, active-unit filtering, and a "reset zoom" functionality for enhanced data discovery.
- **Backend & Security:** Established high-performance database connections utilizing SQLAlchemy and PyODBC, while managing sensitive credentials securely through .env and .gitignore practices.
- **UX Optimization:** Integrated Watchdog for performance monitoring and implemented detailed hover-over metadata displays to provide granular information without cluttering the main visual.
- **Tools Used:** Python 3.13, Streamlit, Plotly Express, Pandas, SQLAlchemy, PyODBC, Watchdog.

8) **University Publication Analysis Dashboard** | Data Scientist & Developer

Internship Project – Çukurova University

- **Project Overview:** Engineered a Streamlit-based web application designed to analyze and visualize academic publication data, providing users with the ability to generate real-time insights from SQL Server or offline CSV datasets.
- **Data Integration:** Built a flexible data sourcing module that executes real-time queries from Microsoft SQL Server while supporting fallback CSV uploads for offline system validation and testing.
- **Analytical Modeling:** Implemented K-Means Clustering to categorize similar academic units and utilized Facebook Prophet for long-term forecasting of future publication trends.
- **Interactive Visualization:** Designed multi-dimensional dashboards featuring line charts, bar graphs, and treemaps for comparative analysis across different academic fields and departments.
- **Granular Exploration:** Developed dynamic filtering systems that allow users to drill down into trends by specific year, research area, or academic unit for multi-level business intelligence.

- Tools Used: Python, Streamlit, MS SQL Server, K-Means (Scikit-Learn), Facebook Prophet, Pandas, Data Visualization (Charts & Treemaps).

9) **Academic Publication Forecasting & Hierarchy Visualizer** | Data Scientist & Developer

Internship Project – Çukurova University

- Project Overview: Developed a Streamlit-based web application for Çukurova University to provide integrated academic publication analysis, forecasting, and organizational unit hierarchy mapping.
- Hierarchical Exploration: Engineered a dynamic selection system for navigating complex academic structures, enabling users to drill down from faculties to specific departments and sub-units based on YÖKSİS IDs.
- Forecasting Engine: Implemented predictive modeling modules utilizing Facebook Prophet and Linear Regression to generate future publication estimates based on historical data trends.
- Visual Analytics: Designed interactive treemap visualizations and yearly trend charts to simplify the analysis of publication densities across various academic fields.
- Functional Flexibility: Integrated customizable year-range selections and the option for single-year predictions, supporting both root-level and sub-unit inclusive data aggregation.
- Tools Used: Python, Streamlit, SQL Server, Facebook Prophet, Linear Regression, Treemap Visualization.

Work Samples

1) **Online KIBHAS Ticket Reservation System** | Database Developer

Academic Project – Software Planning and Management | March 2025 – April 2025

- Project Overview: Architected a comprehensive software project plan and architectural design for a web-based bus reservation system aimed at digitizing transportation services in Cyprus.
- Database Design & Architecture: Designed a relational database schema and Entity-Relationship (ER) diagrams tailored for high-frequency transactions, including seat availability, passenger records, and automated booking confirmations.
- Query & Logic Development: Modeled data classification structures and planned optimized SQL query modules for real-time seat selection and dynamic ticket pricing.
- Technical Planning: Conducted technical feasibility studies for integrating PostgreSQL with a Node.js backend, ensuring data integrity and scalability.
- Security & Compliance: Defined data architecture standards following GDPR for personal data protection and PCI-DSS for secure payment processing simulations.
- SDLC & Methodology: Utilized Agile (Scrum) methodology and the Evolutionary Development Model to manage the project lifecycle, from requirement analysis to testing phase planning.
- Tools Used: MS Visio (System Modeling), MS Project (Resource Scheduling), Figma (UI/UX workflow), and Modelio (UML).

2) **Dynamic Weather Prediction & Smart Farming System** | System Architect

Academic Project – Software Engineering Management | October 2025 – December 2025

- Project Overview: Designed a smart farming decision-making assistant that utilizes real-time weather forecasting and IoT sensor data to provide actionable recommendations for irrigation and planting.

- **System Architecture:** Architected a layered modular architecture comprising a Data Collection Layer, AI Processing Core, and a Physical Field Layer for autonomous control.
- **Technical Solutions:** Designed a "Data Processing Layer" middleware to resolve data normalization challenges by standardizing disparate API and sensor inputs into a unified JSON format.
- **IoT & Automation Logic:** Modeled a smart irrigation loop featuring soil moisture analysis and "Virtual Actuators" to simulate automated valve control within a 5-second hardware response constraint.
- **Tools Used:** UML (Use Case & Class Diagrams), API Gateway, IoT Sensor Simulation, and Map-based Field Management interfaces.

3) **Community Event Management System for TRNC** | Project Manager / Developer

Academic Project – Software Project Planning and Management | September 2025 – January 2026

- **Project Overview:** Architected a comprehensive software project plan for a centralized web-based platform to manage and publish municipal events (festivals, workshops, etc.) across TRNC.
- **Analytical Planning:** Conducted root-cause analysis using Fishbone (Ishikawa) Diagrams and prioritized user pain points through Pareto Analysis to define core system requirements.
- **Feasibility & Estimation:** Performed rigorous effort estimation using the COCOMO model (67.5 person-months) and established project timelines via PERT/CPM critical path analysis.
- **SDLC & Quality Assurance:** Utilized Agile (Scrum) methodology for sprint planning and defined a Quality Function Deployment (QFD) matrix to map user needs to technical design features.
- **Tools Used:** MS Project (Gantt Charting & Crashing), Figma (UI/UX Mockups), MS Visio, and Python-based system modeling.

4) **Multi-Modal Biometric Authentication System** | Computer Vision & Liveness Detection Developer

Graduation Project (4-Person Team) – Software Engineering | October 2025 – June 2026

- **Project Overview:** Designed a modular, prototype-stage biometric authentication platform combining 1:N face identification, voice verification, and anti-spoofing, achieving under 3-second authentication with error-handling simulation modes.
- **Computer Vision & Liveness:** Developed real-time facial landmark analysis (eye-blink, head-pose) and image quality checks (pose, eyes, size, blur). Synchronized audio with lip-movement tracking to prevent pre-recorded video spoofing attacks.
- **System Architecture & Security:** Engineered a REST-based FastAPI and Vanilla JavaScript architecture. Implemented JWT, experimental X-Client flows, and security questions, coordinated by an AuthenticationService tracking actions in an AuditLog.
- **Biometric Fusion & Database:** Extracted embeddings from 15 face and 10 voice samples for a 55/45 weighted fusion score (0.80 threshold), targeting <2% FAR while tracking FRR. Stored encrypted template hashes via async SQLAlchemy in a Supabase PostgreSQL database, enforcing PK/FK constraints across USERS, BIOMETRIC_DATA, SECURITY_QUESTIONS, USER_SECURITY_ANSWERS, and LOGS tables.
- **Tools Used:** Python (3.10/3.11), FastAPI, Uvicorn, HTML/CSS/Vanilla JavaScript, Browser getUserMedia API, PostgreSQL (Supabase), SQLite, async SQLAlchemy, JWT, InsightFace, OpenCV, MediaPipe (FaceMesh), AASIST, Librosa, SoundFile, Resemblyzer, PyTorch, ONNX Runtime, ffmpeg, libsndfile.

5) **Palmora Hotel Reservation System** | Database Administrator & QA Lead

Academic Project – CMSE 322 Software Design | March 2026 – May 2026

- **Project Overview:** Designed a full-stack hotel management platform to streamline operations and improve guest booking using Clean Code and KISS principles.
- **System Architecture:** Engineered a 4-layer architecture featuring dynamic state management and responsive UI/UX for seamless screen confirmation workflows.
- **Technical Solutions:** Implemented double-booking prevention algorithms leveraging B-Tree indexing, Serializable Isolation, row-level locking, and ACID compliance to maintain real-time room availability.
- **Security & Integrations:** Applied OWASP, GDPR/KVKK, and PCI-DSS standards; implemented JWT, 256-bit SSL/TLS, and RBAC. Integrated Stripe API, SendGrid, and Twilio for secure payments and automated notifications.
- **SDLC & QA:** Executed Agile (Scrum) with COCOMO effort estimation and PERT/CPM critical path analysis, authoring SRS, SDS, and PPM reports alongside unit (>80%) and integration testing.
- **Tools Used:** HTML5/CSS3/JavaScript, React, Vite, Node.js, Express.js, MS SQL Server, SQLite, MS Visio, MS Project, Figma, Modelio/draw.io (UML & Gane-Sarson DFDs), Git/GitHub (Git Flow), ESLint, VS Code.

6) **IoT Infant Incubator with Node-RED Dashboard** | High-End System Developer

Academic Project – CMPE 420 & CMSE 423 Embedded System Design | September 2025 – December 2025

- **Project Overview:** Implemented an IoT-enabled Cyber-Physical System (CPS) for an infant incubator featuring real-time environmental sensing, closed-loop control, and web telemetry.
- **Embedded Logic & Automation:** Programmed a microcontroller with a P-Control algorithm and Finite State Machine (FSM) logic to drive PWM heater actuation, utilizing 10-bit ADC processing for thermistors calibrated via two-point linear approximation.
- **Serial Communication:** Developed event-driven and periodic UART/Serial protocols (9600 bps, 8N1) ensuring packet validation and synchronization between the low-end controller and high-end dashboard.
- **Telemetry & Simulation:** Built interactive Node-RED web dashboard flows on a Raspberry Pi for remote monitoring of telemetry, alarms, and control parameters.
- **Tools Used:** Arduino UNO/ATmega (C/C++), Arduino IDE, Node-RED, Scilab/XCOS/Matlab (Thermal Dynamics), Proteus ISIS (Hardware/LCD Simulation), VSPE, 10k Ω NTC thermistors, 6.2k Ω resistors, LM016L 16x2 LCD.

7) **Online Image Encryption & Digital Signature Web Application** | Full-Stack Developer

Academic Project – CMSE 456 & CMPE 455 Security and Cryptography | February 2026 – May 2026

- **Project Overview:** Architected a full-stack secure web application for encrypted image transmission, digital signature verification, and log auditing over a LAN.
- **System Architecture:** Built Client and Sysadmin interfaces connecting to a high-performance backend utilizing robust RESTful API routing and database integration.
- **Cryptography Implementations:** Developed 2048-bit RSA key exchange with automated background refresh for forward secrecy, and encrypted binary images using DES-CBC mode with random 8-byte IVs and PKCS#7 padding.

- **Payload & Signatures:** Managed dynamic array transformations (Uint8Array, Blob) and Base64 stream encoding/decoding. Secured DES session keys with RSA (PKCS#1 v1.5) and applied SHA-256 hashing for digital signatures to ensure data integrity.
- **Tools Used:** Python, FastAPI, Uvicorn ASGI, SQLite, PyCryptodome, HTML5/CSS3/JavaScript, Fetch API, Local Storage API.

8) 101evler Real Estate Portal - Software Quality Assurance & Testing | QA Engineer

Academic Project – CMSE 326 Software Quality Assurance and Testing | March 2026 – June 2026

- **Project Overview:** Conducted multi-layered quality assurance auditing and black-box testing on Northern Cyprus's primary real estate platform (101evler.com).
- **Testing Methodologies:** Executed Manual Black-Box and Exploratory Testing to validate core search algorithms, currency matching, form validations, and navigation workflows.
- **Compatibility & Performance:** Evaluated cross-browser compatibility and simulated mobile environments (iPhone/iPad) to identify layout anomalies. Analyzed performance bottlenecks (e.g., 16.1s LCP, FCP, Speed Index) to formulate lazy loading and script deferral optimizations.
- **Analytical Reporting:** Conducted asset integrity audits and automated broken link testing, benchmarking Human QA defect severity reports against Generative AI assessments.
- **Tools Used:** Chrome DevTools, Safari Responsive Design Mode, Google PageSpeed Insights, ChatGPT, Gemini, Claude, Grammarly.

Research and Reports

Research Report: “5G Wireless Technology” (2024)

- Explored the benefits, applications, and impact of 5G technology on daily life.
- Analyzed key aspects such as higher speeds, bandwidth, reliability, and real-world applications like streaming, gaming, and IoT.
- Targeted for engineering students, technology professionals, and enthusiasts

Certifications

- Explore Emerging Tech, IBM
- Coding with IBM: Cyberstart, Kodluyoruz
- GIT & GITHUB, Eastern Mediterranean University
- Career and Coding, Eastern Mediterranean University
- Artificial Intelligence and Data, Eastern Mediterranean University
- Artificial Intelligence Summit'24, Eastern Mediterranean University
- Woman in Technology, Eastern Mediterranean University
- Lexathon'23 Legal Tech Development Area, Turkcell
- Data Science and Artificial Intelligence. Turkcell Geleceği Yazarlar
- The Impact of Data Science and Artificial Intelligence on Our Future and Career Goals. EMU

Awards

- High School Valedictorian Plaque, Private Adana Ceyhan Final Anatolian High School (2022)